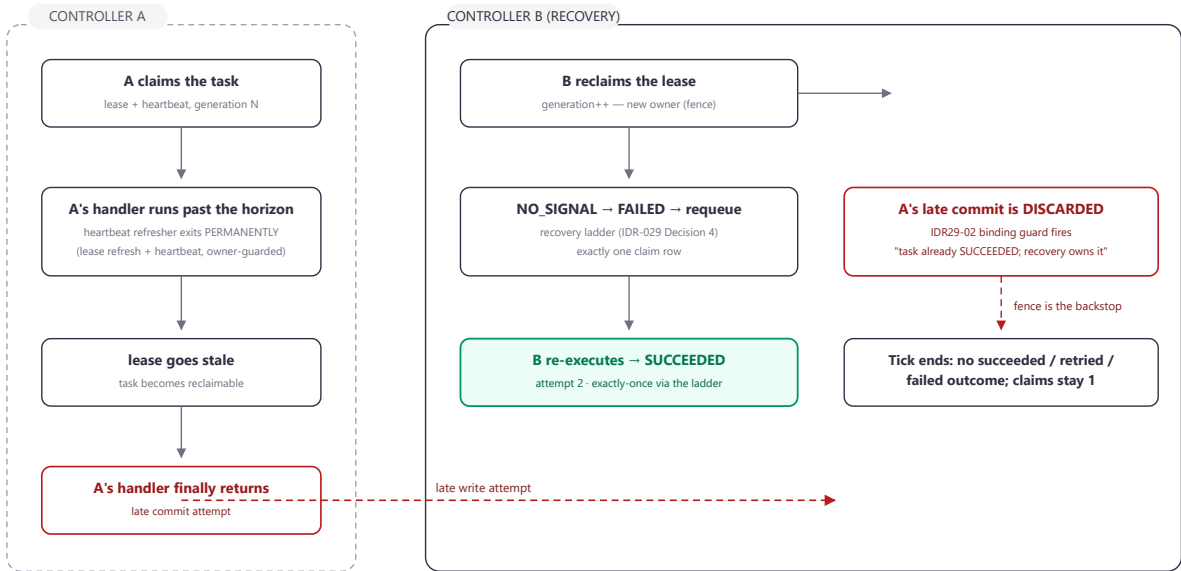


Adversarial review — the heartbeat-refresh horizon cliff

Semantic pattern: **sequential recovery with stale-write rejection** → visual type: **flowchart** · a handler longer than the refresh horizon · verified at HEAD `ac770f7`



The binding guard fires BEFORE the generation fence: the acceptance path's status read sees B's terminal SUCCEEDED and refuses — the fence

Provenance manifest — diagram source

Item	Value
Repository commit	Hermes <code>ac770f7</code> (clean tree)
Semantic pattern → type	Sequential recovery with stale-write rejection → Flowchart (harvested from <code>cathrynlavery/diagram-design v2.4 @ 09df49d8</code> , MIT)
Files inspected	<code>src/hermes/research/controller.py</code> (<code>_refresh_liveness</code> — lease + heartbeat owner-guarded, horizon-exhaustion permanent exit, <code>NO_SIGNAL</code> ladder, IDR29-02 binding guard, generation fence), <code>src/hermes/research/controller.py</code> (<code>DEFAULT_HEARTBEAT_REFRESH_HORIZON = 300.0</code>)
Architecture documents	<code>hermes_p2_residuals_governance.md</code> (P2 #4 accepted design), IDR-029 (Decision 4 recovery), verdict record (sharp probe at post-F15 HEAD)

Tests	test_15_f15_horizon_cliff_late_handler_discarded (A-side: late return discarded, claims stay 1), test_15_f15_hung_worker_recovered_after_horizon (B-side: attempt 2, exactly one claim), test_14_f15_live_worker_lease_protected
Node status	All nodes IMPLEMENTED ; the discard path is the tested real order (binding guard before fence)